Let's Play!

The Space Mission Game

Faculty :- Prof Kajal Vatekar

Meet our team:-

Nehali Sawarkar
Smriti Agarwal
Rishika Agarwal
Sonal Bulusu

Overview

Title: Space Mission Game (C Mini Project)

- Console-based game developed in C
- The player acts as an astronaut on a space mission
- Objective: Reach distance 100
- Must manage 3 resources:
  - Fuel
  - Oxygen
  - Health
  - Game difficulty increases with levels

Levels & Difficulty

The game has 3 levels:

- Level 1 (0–49 Distance):
  - Easy problems
- Level 2 (50-84 Distance):
  - Dangerous situations
- Level 3 (85-100 Distance):
  - Critical emergencies

Game Logic & Objective

Working Principle:

- The game starts with:
  - Fuel = 100
  - Oxygen = 100
  - Health = 100
  - Distance = 0
- Random space events occur using rand()
- Player selects 1 of 3 decisions
- Each decision affects resources differently
- Game ends when:
  - Distance >= 100 → Mission Success
  - Distance < 100 and parameters = 0 → Mission Failed

Platform used

VS Code

Libraries used

stdio.h – Used for input and output operations
stdlib.h – Used for random number generation and utility functions
time.h – Used to get the current time for generating random events

Functions Used

- instructions()
  - Displays game instructions.
- applyEffects()
  - Uses pointers
  - Modifies fuel, oxygen, health, and distance
  - Decision-based resource changes
- while loop controls game flow
- Conditional statements manage levels
- playGame()
  - Core game logic
  - Random event generation
  - Level control
  - Passive survival drain
- showResult()
  - Displays the final result.

Conclusion

The Space Mission Game is an interactive C program where the player manages fuel, oxygen, and health to reach 100% distance. It uses basic concepts like loops, switch cases, and random functions to create different events. The player can win by reaching the destination with resources remaining, or lose if any resource becomes zero. This program helps in understanding logic building and game development using C.

Thank You For Playing!